

CSE 4810 : Algorithm Engineering Lab

Lab - 2

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Task 1

Let's say you have a BST and now I want you to find all the available nodes in the given range. First convert the list given in first line of the input to a BST using insert function given in the helping code. Then try to find all the nodes available in the given range in the second line.

Input:

ar = [11, 2, 29, 1, 7, 15, 40, 35]

range = (12,35)

output: 15, 29, 35

Task 2

Given a binary search tree with a value associated with each node, we need to choose a subset of these nodes such that the sum of selected nodes is maximum under a constraint that no two chosen nodes in the subset should be directly connected, that is, if we have taken a node in our sum then we can't take any of its children in consideration and vice versa. Input will be a list of integers as the nodes of the BST and output will be a single integer.

Input: ar = [11, 2, 29, 1, 7, 15, 40, 35]. Output : 74

Helping Code

```
class Node:
    def __init__(self, x):
        self.data = x
        self.left = None
        self.right = None
def insert(root, key):
    if root is None: return Node(key)
    if root.data == key: return root
    if root.data < key: root.right = insert(root.right, key)
    else: root.left = insert(root.left, key)
    return root
if __name__ == '__main__':
    ar, root = [11, 2, 29, 1, 7, 15, 40, 35], None
    for el in ar:
        root = insert(root, el)
    print(root.data)
```